



Inspiring Excellence

CSE370 : Database Systems

Project Report

Project Title : Water Management System

Group No : 02 , CSE370 Lab Section : 17 , Spring 2026		
ID	Name	Contribution
24101439	Md Moniruzzaman	Frontend,Backend
24341025	Raiyan Mahamud	Frontend,Backend
24101096	MD. Atique-Ur-Rahman Bhuiyan	Frontend,Backend

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Introduction

Water Bill Management System : a web-based utility management platform for water supply authorities in Bangladesh. It automates the full billing lifecycle: admins enter meter readings and the system calculates bills automatically ($\text{consumption} \times \text{rate} + \text{tax}$), applies subsidies for low-income customers (under ₳30,000/month), and adds penalties to overbills in bulk. The system also acts as a smart monitoring tool, it auto-detects abnormal usage (possible leakage or illegal connections) and faulty meters, raising real-time alerts visible to both admins and customers. Admins can manage meter status, filter outstanding dues, and take bulk actions, while customers only see their own data. Everything is protected by role-based access control with separate portals for admins and customers.

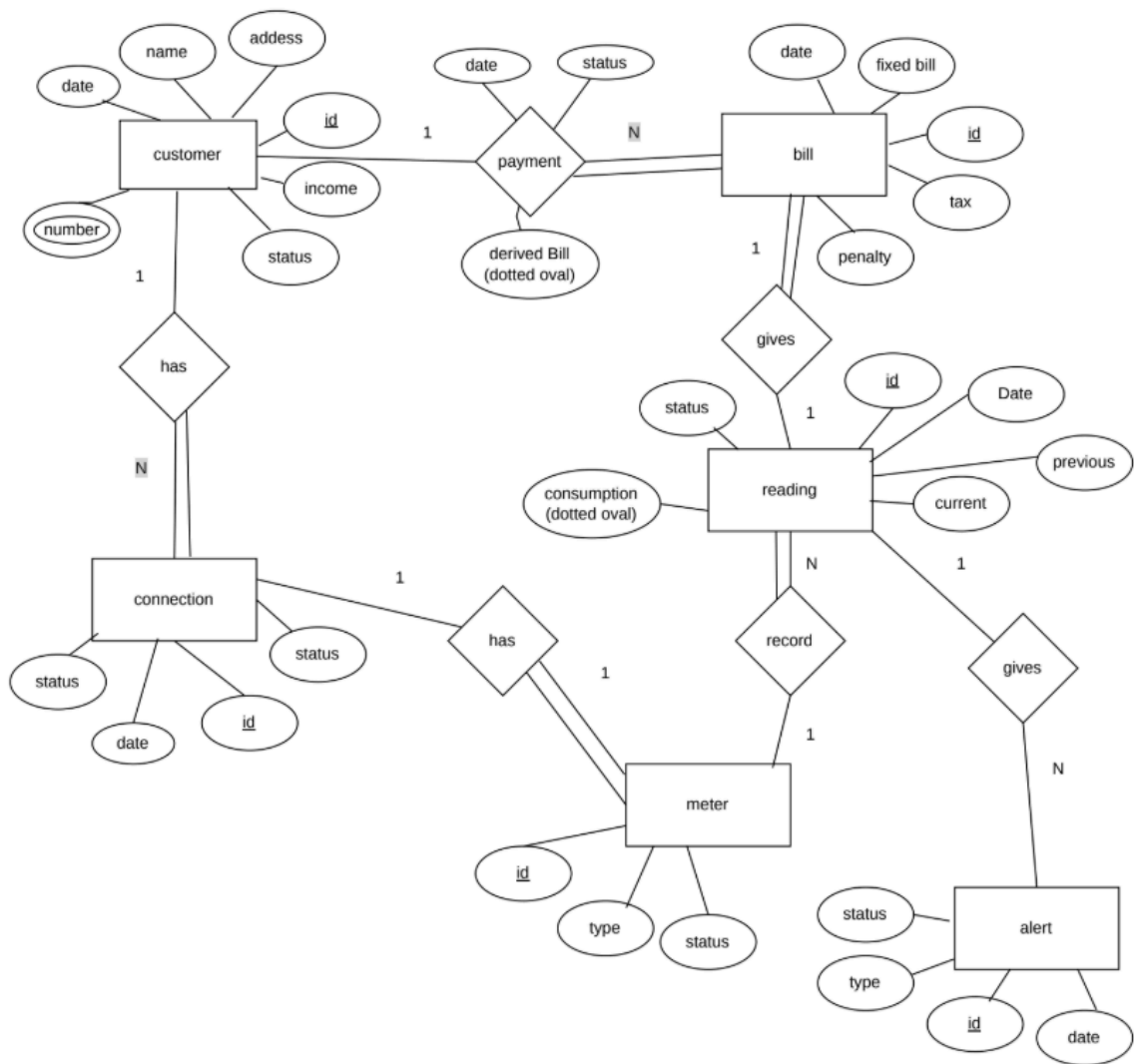
In short: it replaces manual water billing work with an automated, fair, and alert-driven system tailored for a Bangladeshi utility context.

Project Features

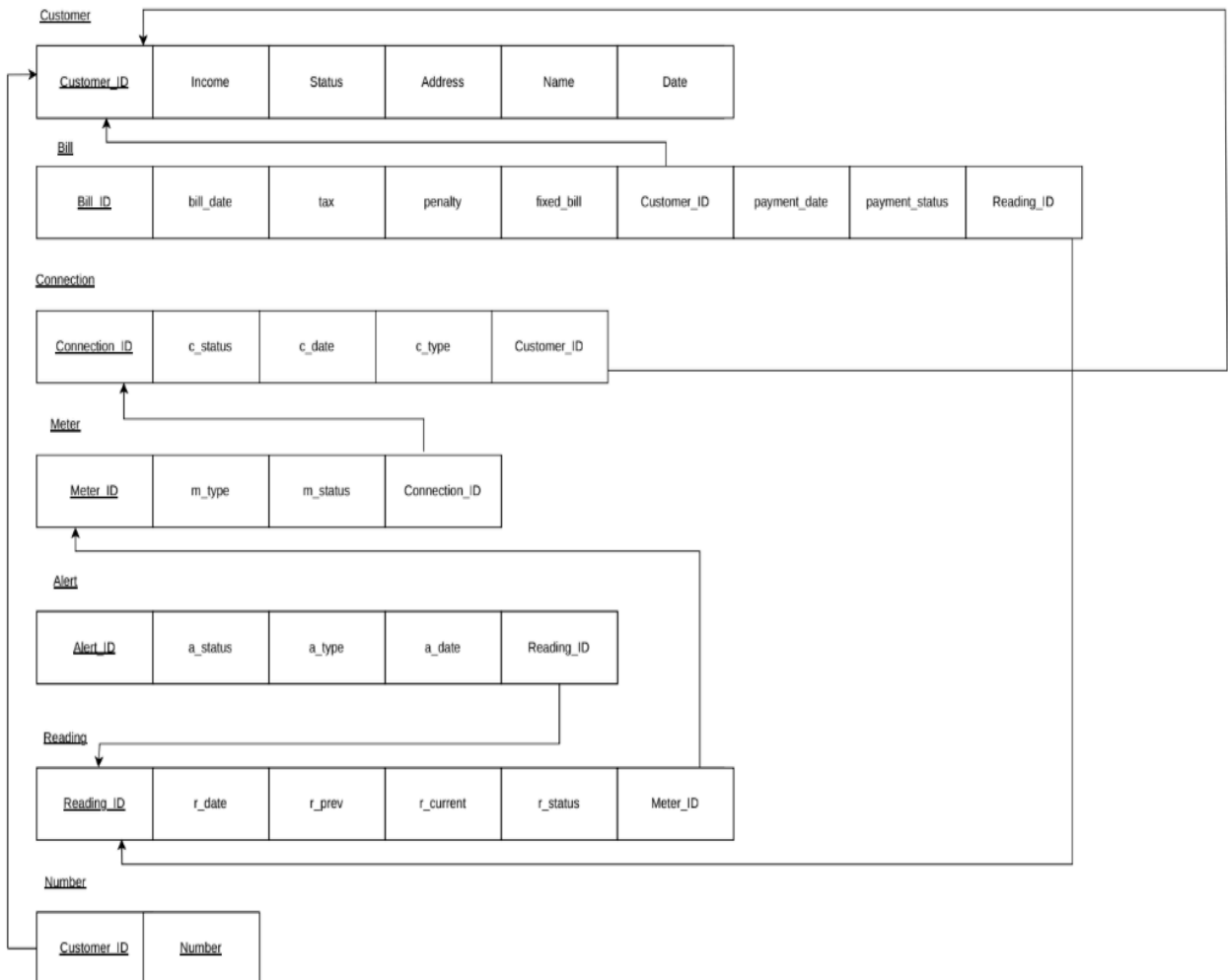
ID, Name	Features [3 per member]	
24101439, Md Moniruzzaman	Ft 1	Automatic Bill Generation System auto-calculates bills when admin adds reading. No manual math needed. $\text{consumption} \times \text{rate} + \text{tax} = \text{bill}$. All done by code.
	Ft 2	Faulty Meter Detection If meter reading is the same as last month (zero consumption) → system flag as Faulty and create alert. Admin knows the meter is broken without checking manually.
	Ft 3	Automatic Penalty System Admin click one button → all unpaid bills older than 30 days marked Overdue + ₳50 penalty added each. No need to check one by one.
24341025, Raiyan Mahamud	Ft 1	Subsidy for Low-Income Customers If customer income below ₳30,000 → system auto give 10% discount on every bill. Fair system for poor people.
	Ft 2	Real-Time Alert System When reading is abnormal or faulty → alert created instantly in database. Admin sees it in the dashboard. Customers see warning banners. Both were notified at the same time.

	Ft 3	Role-Based Access Control Two types of user — Customer and Admin. Customers see only their own data. Admin sees everything. Different login page, different dashboard, different features. Data stay safe.
24101096, MD. Atique-Ur- Rahman Bhuiyan	Ft 1	Leakage & Abnormal Usage Detection If water usage suddenly is very high (3× average) → system auto create Leakage alert. Helps find illegal connections or pipes burst fast.
	Ft 2	Meter Activate / Deactivate Admin can turn off any customer meter with one click. System update both meter table AND connection table together using transaction. No incomplete update.
	Ft 3	Outstanding Dues Query System Admin can filter all unpaid bills by status, minimum amount, or customer name. See per-customer total debt. Useful for collecting overdue payments fast.

ER Diagram



Schema Diagram



Normalization

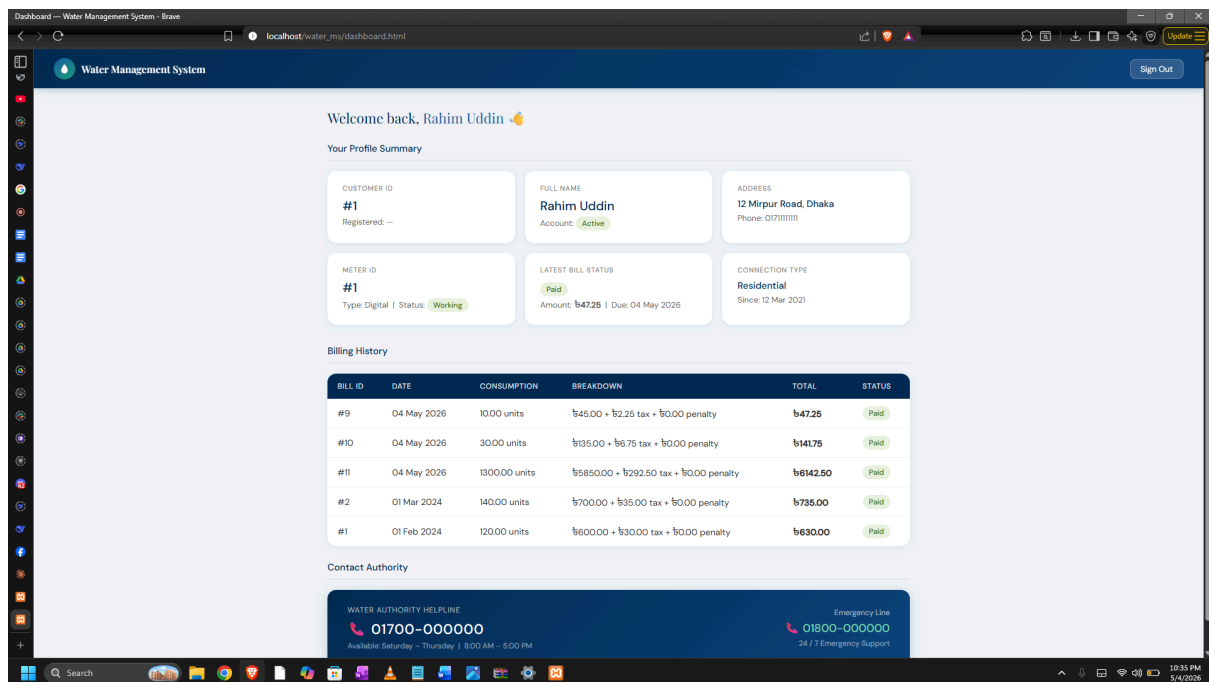
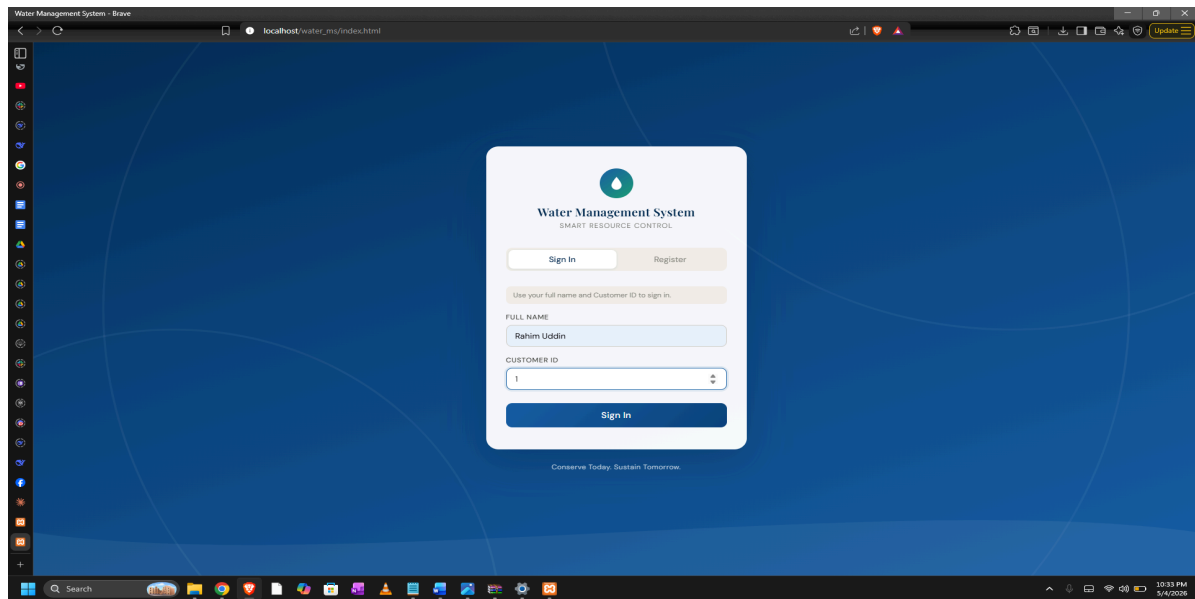
- a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.
- b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?
- c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

Ans: It is already in 3NF form, so it has already surpassed 1NF and 2NF form.

Frontend Development

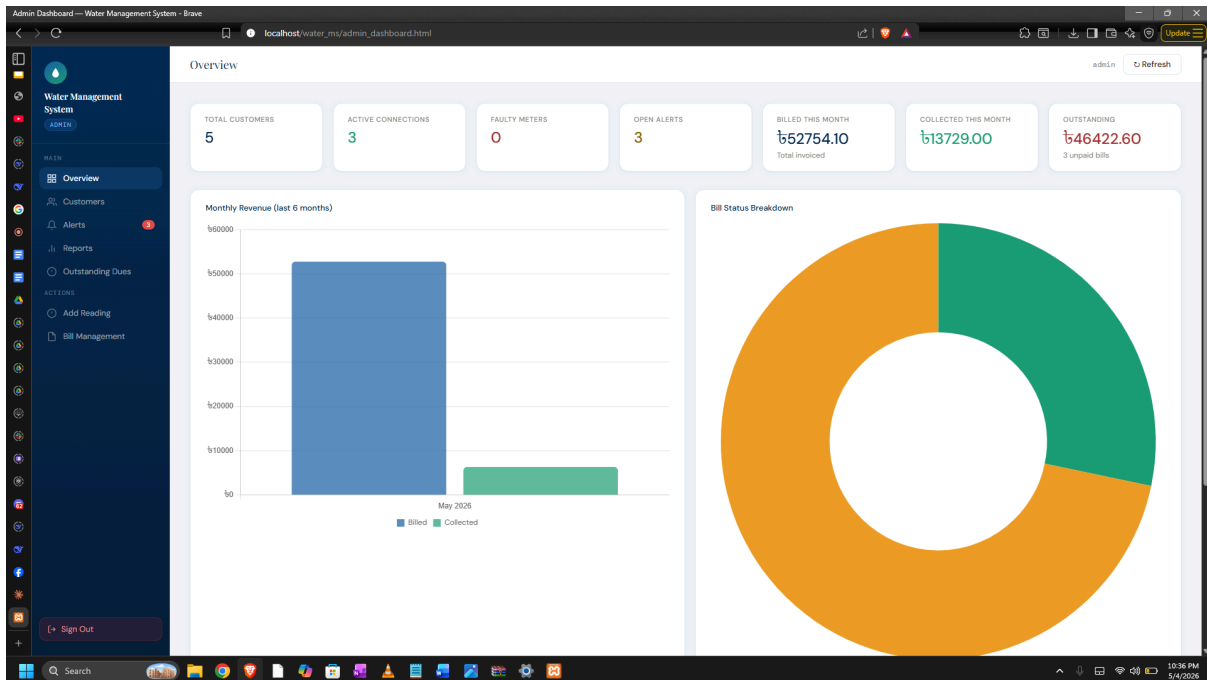
Contribution of ID : 24101439, Name : Md Moniruzzaman

Monir made the frontend User login and register page along with the dashboard of user using HTML, CSS and Javascript



Contribution of ID : 24341025, Name : Raiyan Mahamud

Raiyan made most of the frontend of the admin page as he did Admin login along with Admin dashboard and Bill Management page using HTML, CSS and Javascript



The screenshot displays the Admin Login page for the Water Management System. The page has a dark blue background with a central login form. The form includes a header with the system name and a note that access is restricted to authorized personnel. It provides default credentials (Username: admin, Password: admin123) and fields for the user to enter their Admin Username and Password. A 'Sign in to Admin Panel' button is located below the password field, and a link to 'Back to customer login' is at the bottom of the form.

Water Management System
Restricted — authorised personnel only

Default credentials:
Username: admin | Password: admin123

ADMIN USERNAME
admin

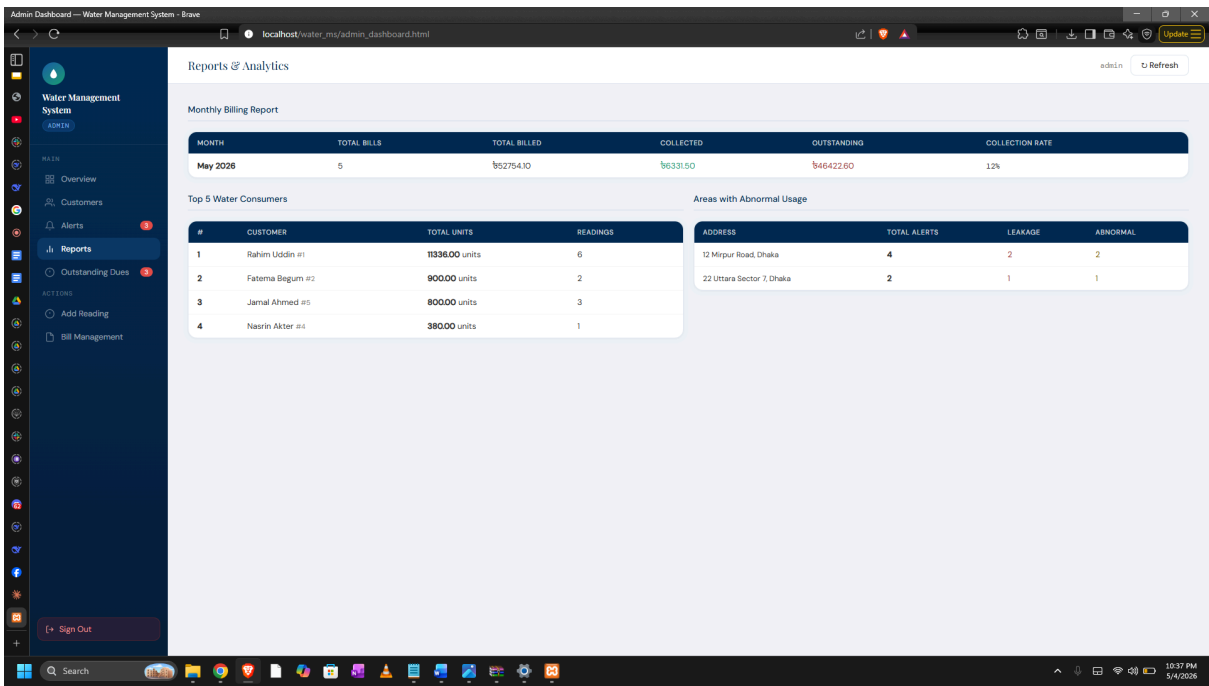
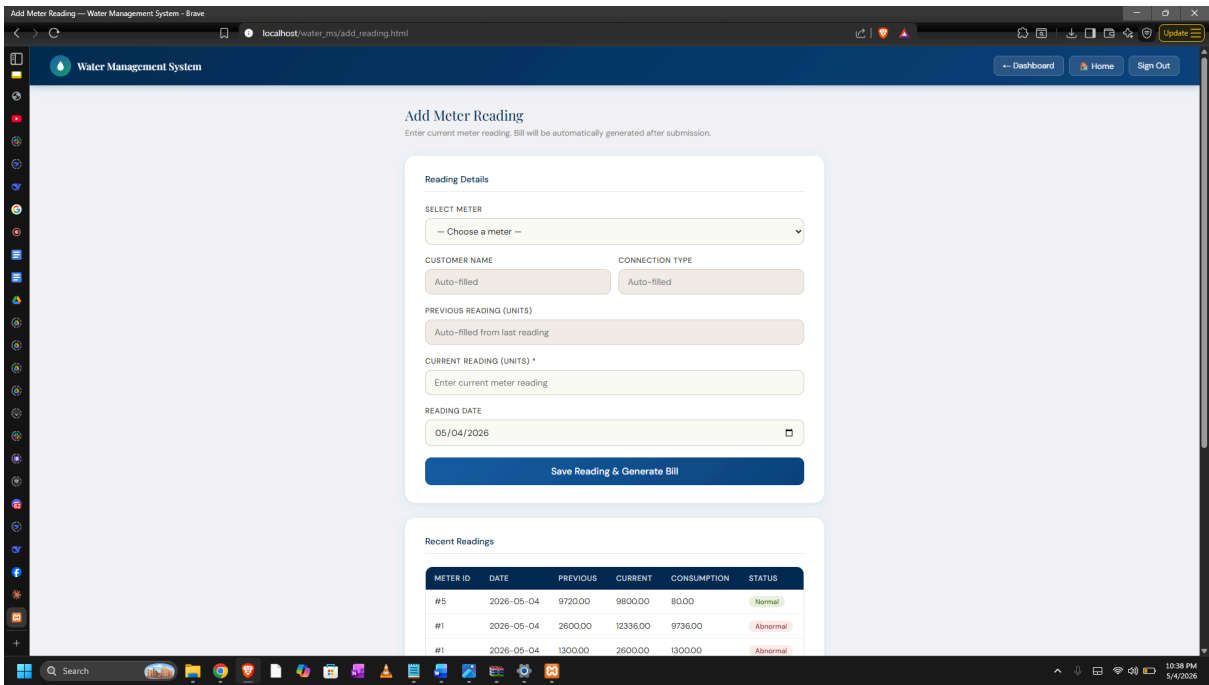
PASSWORD

Sign in to Admin Panel

[Back to customer login](#)

Contribution of ID : 24101096, Name : MD. Atique-Ur-Rahman Bhuiyan

Atique made the part of Reports and Add Meter of the admin page using HTML, CSS and Javascript



Backend Development

Contribution of ID : 24101439, Name : Md Moniruzzaman

Monir made the part of Admin Reading and And Generate Bills using PHP

[illegible][illegible]

Contribution of ID : 24341025, Name : Raiyan Mahamud

Raiyan made the part of SQL and Admin Data using PHP

```
File Edit Selection View Go Run Terminal Help
E:\> cd c:\hidocs\water_ms > wms_complete.sql

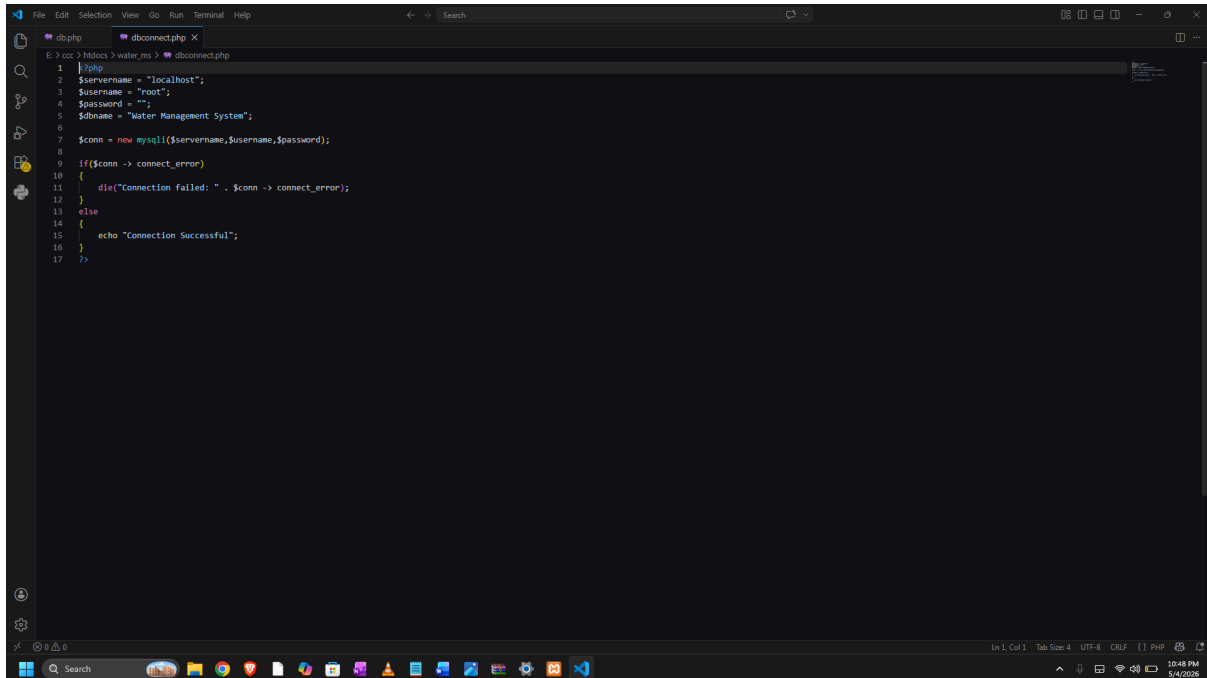
20 CREATE TABLE Connection (
21     Connection_ID INT PRIMARY KEY AUTO_INCREMENT,
22     Customer_ID INT NOT NULL,
23     c_type VARCHAR(30) NOT NULL DEFAULT 'Residential', -- Residential / Commercial
24     c_status VARCHAR(20) NOT NULL DEFAULT 'Active', -- Active / Inactive
25     c_date DATE NOT NULL DEFAULT (CURRENT_DATE),
26     CONSTRAINT fk_conn_customer FOREIGN KEY (Customer_ID)
27     REFERENCES Customer(Customer_ID) ON DELETE CASCADE
28 );
29
30 CREATE TABLE Meter (
31     Meter_ID INT PRIMARY KEY AUTO_INCREMENT,
32     Connection_ID INT NOT NULL,
33     m_type VARCHAR(30) NOT NULL DEFAULT 'Digital', -- Digital / Analog
34     m_status VARCHAR(20) NOT NULL DEFAULT 'Working', -- Working / Faulty / Missing
35     CONSTRAINT fk_meter_conn FOREIGN KEY (Connection_ID)
36     REFERENCES Connection(Connection_ID) ON DELETE CASCADE
37 );
38
39 CREATE TABLE Reading (
40     Reading_ID INT PRIMARY KEY AUTO_INCREMENT,
41     Meter_ID INT NOT NULL,
42     r_date DATE NOT NULL DEFAULT (CURRENT_DATE),
43     r_prev DECIMAL(10,2) NOT NULL DEFAULT 0.00, -- previous meter reading
44     r_current DECIMAL(10,2) NOT NULL DEFAULT 0.00, -- current meter reading
45     r_status VARCHAR(20) NOT NULL DEFAULT 'Normal', -- Normal / Abnormal / Faulty
46     CONSTRAINT fk_reading_meter FOREIGN KEY (Meter_ID)
47     REFERENCES Meter(Meter_ID) ON DELETE CASCADE
48 );
49
50 CREATE TABLE Bill (
51     Bill_ID INT PRIMARY KEY AUTO_INCREMENT,
52     Customer_ID INT NOT NULL,
53     Reading_ID INT NOT NULL,
54     bill_date DATE NOT NULL DEFAULT (CURRENT_DATE),
55     fixed_bill DECIMAL(10,2) NOT NULL DEFAULT 0.00, -- base charge
56     tax DECIMAL(10,2) NOT NULL DEFAULT 0.00,
57     penalty DECIMAL(10,2) NOT NULL DEFAULT 0.00,
58     payment_date DATE,
59     payment_status VARCHAR(20) NOT NULL DEFAULT 'Unpaid', -- Paid / Unpaid / Overdue
60     CONSTRAINT fk_bill_customer FOREIGN KEY (Customer_ID)
61     REFERENCES Customer(Customer_ID) ON DELETE CASCADE,
62     CONSTRAINT fk_bill_reading FOREIGN KEY (Reading_ID)
63     REFERENCES Reading(Reading_ID) ON DELETE CASCADE
64 );
65
66 CREATE TABLE Number (
67     Number_ID INT PRIMARY KEY AUTO_INCREMENT,
68     Customer_ID INT NOT NULL,
69     Number VARCHAR(20) NOT NULL,
70     CONSTRAINT fk_number_customer FOREIGN KEY (Customer_ID)
71     REFERENCES Customer(Customer_ID) ON DELETE CASCADE
72 );
```

```
File Edit Selection View Go Run Terminal Help
E:\> cd c:\hidocs\water_ms > admin_data.php

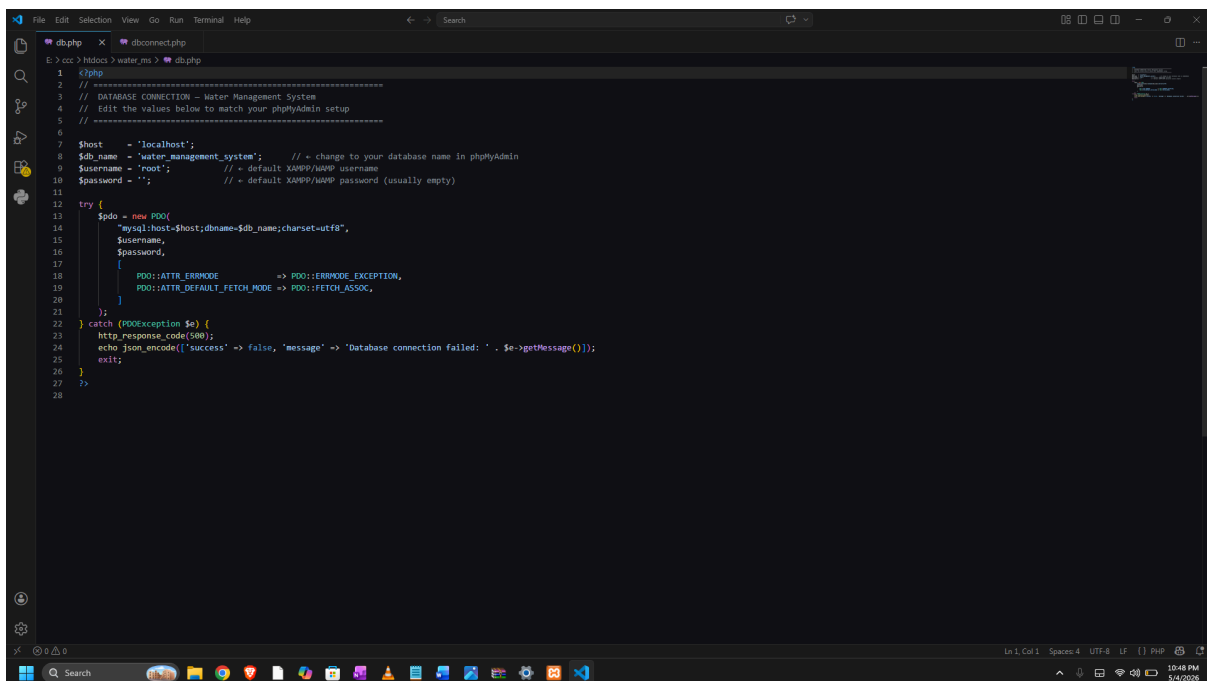
85 // =====
86 // ACTION: ALL OPEN ALERTS
87 // =====
88 if ($action == 'get_alerts') {
89     $stmt = $pdo->query(
90         "SELECT
91             a.Alert_ID, a.a_type, a.a_status, a.a_date,
92             r.Reading_ID, r.r_prev, r.r_current,
93             (r.r_current - r.r_prev) AS consumption,
94             r.r_status,
95             m.Meter_ID, m.m_status,
96             cu.Customer_ID, cu.Name AS customer_name, cu.Address
97         FROM alert a
98         JOIN reading r ON a.Reading_ID = r.Reading_ID
99         JOIN meter m ON r.Meter_ID = m.Meter_ID
100         JOIN connection cn ON m.Connection_ID = cn.Connection_ID
101         JOIN customer cu ON cn.Customer_ID = cu.Customer_ID
102         WHERE a.a_status = 'Open'
103         ORDER BY a.a_date DESC"
104     );
105     $alerts = $stmt->fetchAll();
106     echo json_encode(['success' => true, 'alerts' => $alerts]);
107     exit;
108 }
109
110 // =====
111 // ACTION: RESOLVE ALERT
112 // =====
113 if ($action == 'resolve_alert') {
114     $alert_id = intval($data['alert_id']) ?? 0;
115     if (!$alert_id) { echo json_encode(['success' => false, 'message' => 'Invalid alert ID.']); exit; }
116     $stmt = $pdo->prepare("UPDATE alert SET a_status='Resolved' WHERE Alert_ID=?");
117     $stmt->execute([$alert_id]);
118     echo json_encode(['success' => true, 'message' => "Alert $alert_id resolved."]);
119     exit;
120 }
121
122 // =====
123 // ACTION: MONTHLY REPORT
124 // =====
125 if ($action == 'get_reports') {
126     // Monthly billing trend (last 6 months)
127     $monthly = $pdo->query(
128         "SELECT
129             DATE_FORMAT(bill_date, '%Y %m') AS month,
130             SUM(fixed_bill + tax + penalty) AS total_bill"
131     );
132     $reports = $monthly->fetchAll();
133     echo json_encode(['success' => true, 'reports' => $reports]);
134     exit;
135 }
```

Contribution of ID : 24101096, Name : MD. Atique-Ur-Rahman Bhuiyan

Atique connected the PHP and SQL using DB and DB connect



```
1 <?php
2 $servername = "localhost";
3 $username = "root";
4 $password = "";
5 $dbname = "Water Management System";
6
7 $conn = new mysqli($servername,$username,$password);
8
9 if($conn -> connect_error)
10 {
11     die("Connection failed: " . $conn -> connect_error);
12 }
13 else
14 {
15     echo "Connection Successful";
16 }
17 ?>
```



```
1 <?php
2 // =====
3 // DATABASE CONNECTION - Water Management System
4 // Edit the values below to match your phpMyAdmin setup
5 // =====
6
7 $host = 'localhost';
8 $db_name = 'water_management_system'; // + change to your database name in phpMyAdmin
9 $username = 'root'; // + default XAMPP/WAMP username
10 $password = ''; // + default XAMPP/WAMP password (usually empty)
11
12 try {
13     $pdo = new PDO(
14         "mysql:host=$host;dbname=$db_name;charset=utf8",
15         $username,
16         $password,
17         [
18             PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
19             PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
20         ]
21     );
22 } catch (PDOException $e) {
23     http_response_code(500);
24     echo json_encode(['success' => false, 'message' => 'Database connection failed: ' . $e->getMessage()]);
25     exit;
26 }
27 ?>
```

Source Code Repository

<https://github.com/MdMoniruzzaman433/Md-Monir>

Conclusion

This project is a water billing system for managing customers and their water usage. It can generate bills automatically, send alerts for problems, and let admins manage everything easily. We learned how to connect a database with a website using PHP and MySQL. Overall the project was a success. We learned a lot from this Project

References

<https://youtu.be/d5Hf6d6QtIo?si=7IcIjT4HDdXdb0Yd>

https://youtu.be/J8QPm8_j6lM?si=vfDGJMiEIFiey_8t

<https://youtu.be/K-qXW9ymeYQ?si=Etbm2ww9EJQThMdG>